

AJAY KRISHNA DEVULAPALLY

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PROFESSIONAL SUMMARY

Software engineer and M.S. Computer Science student at Penn State with experience building Python and C/C++ systems, backend-oriented workflows, simulation pipelines, and performance-focused software. Strong foundations in data structures, algorithms, operating systems, databases, debugging, testing, and object-oriented design, with hands-on experience in REST APIs, scheduling, memory management, SQL, Linux, Git, AWS concepts, Docker, and machine-learning systems.

EDUCATION

The Pennsylvania State University

M.S. in Computer Science, Thesis Track

CVR College of Engineering

B.Tech. in Computer Science and Engineering

University Park, PA

Aug 2024 – Aug 2026

Hyderabad, India

2019 – 2023

EXPERIENCE

Graduate Research Assistant

Aug 2024 – Present

Microsystems Design Lab, Penn State | SRC

University Park, PA

- Conduct SRC-supported research under **Prof. Vijaykrishnan Narayanan** on AI/ML-driven optimization for microsystems and combinatorial design problems.
- Built Python, C++, and PyTorch workflows integrating simulation, routing optimization, batch execution, data processing, benchmarking, and reproducible experiment configurations.
- Automated evaluation across multiple design and graph instances, debugging cross-component failures and comparing runtime, feasibility, constraint violations, and solution quality.
- Developing GNN-based freeze-mask methods for oscillator-based Ising-machine optimization to reduce runtime while preserving solution quality on unseen graphs.

AI Intern

Mar 2021 – Apr 2021

Smartknower

Bangalore, India

- Built ANN-based and transfer-learning image classification models, achieving 98.3% accuracy on MNIST.
- Implemented preprocessing, validation, evaluation, and Streamlit demonstration workflows using Python, TensorFlow, Keras, OpenCV, and Pandas.

RESEARCH

TWIRLeT-2.5D – First Author

2026

[*TCAD 2026*] *Ajaykrishna Devulapally, Varun Parekh, Anusha Devulapally, Mirco Sciuilli, Srivatsa Rangchar Srinivasa, Vijaykrishnan Narayanan, “TWIRLeT-2.5D: Thermal Wirelength Integrated Reinforcement Learning for 2.5D Chiplet Placement.”*

- Built an end-to-end Python/PyTorch system for constrained chiplet placement using imitation learning, PPO, action masking, MILP-based routing, thermal simulation, and automated benchmark evaluation.
- Reduced wirelength by up to 27% and achieved 58× faster time-to-solution than simulated annealing through reusable inference and evaluation workflows.

PROJECTS

Cloud-Native Data Service | Python, REST APIs, AWS, SQL, Docker

2026

- Designed a modular backend-service prototype with REST endpoints, request validation, SQL-backed metadata storage, structured logging, and separated service and data-access layers.
- Mapped the deployment architecture to AWS S3, RDS/DynamoDB, Lambda, API Gateway, EC2, CloudWatch, and IAM concepts, with Docker and GitHub Actions for repeatable builds and CI/CD-oriented delivery.

Preemptive Thread Scheduler | C, Linux, Operating Systems

2025

- Built a tick-accurate scheduler implementing FCFS, SRTF, and five-level MLFQ with preemption, I/O blocking, priority boosting, deterministic state transitions, and queue management.
- Validated scheduling behavior against expected Gantt traces and debugged edge cases involving blocking, preemption, tie-breaking, and workload transitions.

Buddy and Slab Memory Allocator | C, Operating Systems

2025

- Implemented a custom heap allocator using buddy and slab allocation with chunk splitting, recursive coalescing, free-list management, metadata headers, and deterministic address selection.
- Built `my_malloc` and `my_free` interfaces with correctness-focused tests for allocation, deallocation, merging, small-object reuse, and fragmentation behavior.

SKILLS

Programming: Python, C, C++, Java, SQL, JavaScript, TypeScript

Software Engineering: Object-oriented programming, data structures, algorithms, REST APIs, modular design, debugging, unit and integration testing

Systems / Backend: Linux, operating systems, multithreading, synchronization, CPU scheduling, memory management, relational databases

Cloud / Tools: Git, GitHub, Docker, GitHub Actions, CI/CD concepts, AWS S3, EC2, Lambda, RDS, DynamoDB

AI / ML: PyTorch, TensorFlow, Scikit-learn, model evaluation and experimentation

Certificates: Database Programming with SQL – Oracle(2022), Artificial Intelligence – Smartknower (2021),